EECS E6691 COLUMBIA UNIVERSITY SPRING 2023

Advanced Deep Learning

Course syllabus

Mehmet Kerem Turkcan

Department of Electrical Engineering, Columbia University

January 10, 2023

Instructor	Mehmet Kerem Turkcan	Time	Wednesdays, 10:10 AM to 12:40 PM
Email	mkt2126@columbia.edu	Location	1024 Mudd Building

Course description

This is an advanced course on current topics in deep learning, for students who have already taken an introductory deep learning course. It consists of lectures on state-of-the-art architectural and modeling concepts, assignments, an exam, and a final project, and it addresses topics beyond the first course on deep learning (such as ECBM E4040), with applications of interest to students. In 2023, the main subject of the lectures is **object detection**.

Students entering the course need prior experience with deep learning and neural network architectures, including convolutional neural networks (CNNs), recurrent neural networks (RNNs), long short-term memory networks (LSTMs), and autoencoders. They need a working knowledge of Python and its libraries, Jupyter notebooks, TensorFlow on local machines and on Google Cloud, and GitHub or a similar code hosting service. The course focuses on PyTorch and Google Cloud. Students must be self-sufficient learners who take an active role in classroom activities.

There are three to four coding assignments during the semester, and a multiple-choice midterm exam in the second half of the course. Final projects are documented in a conference-style report, with code deposited in a GitHub repository; the code must be documented and instrumented so that the instructor can run it after downloading the repository. A Google Slides presentation of the project, suitable for a poster session, is also required.

Courseworks	https://courseworks2.columbia.edu/courses/168932
Ed Discussion forum	https://edstem.org/us/courses/35588/discussion/
Course directory	https://doc.sis.columbia.edu/#subj/EECS/E6691-20231-001/
Office hours	After class, by appointment, or by posting questions on the Ed Discussion forum.
Teaching assistant	Chengbo Zang.
TA and CA office hours	Announced on Courseworks.

References and prerequisites

The main resource for the course is *Deep Learning* by Ian Goodfellow, Yoshua Bengio, and Aaron Courville (MIT Press, 2016), available at deeplearningbook.org.

- Machine learning, taken previously or in parallel with this course.
- ECBM E4040 Neural Networks and Deep Learning, or an equivalent university course on neural networks and deep learning taken for academic credit.
- An excellent theoretical background in probability, statistics, and linear algebra.

> **Before you enroll**
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> Students without an adequate theoretical background or previous experience programming deep learning models are strongly advised to drop the class. The instructor requires that students take no more than 12 credits of coursework, including this course and project courses, during the semester in which they take this course.

Grading

Assignments, 30%	Four graded assignments. Each student completes their own coding tasks and questions, using Python, TensorFlow, Jupyter notebooks, and code management tools.
Exam, 30% (week 11)	An in-class multiple-choice exam covering deep learning theory, architecture, and design.
Final project, 40%	One project based on contemporary papers or original ideas, in groups of no more than three, documented in code and a report.

Important dates

Date	**Due**
Feb 5	Assignment 1
Mar 12	Assignment 2
Mar 21	Project proposal
Apr 5	**Midterm exam**
Apr 29	Assignment 3
May 10	**Final project**

Course outline

Week	**Topics**
1	Introduction to the course; history and fundamentals of object detection; evaluation criteria for object detection
2	Conventional object detection methods; segmentation
3	Historical segmentation approaches; fully convolutional networks (FCNs); U-Nets for segmentation
4	R-CNN, Fast R-CNN, and Faster R-CNN; generative pre-training (GPT) architectures
5	UNet++, ReInspect, the Hungarian algorithm, multi-task network cascades, feature pyramid networks (FPN), Mask R-CNN, PointRend, Cascade R-CNN
6	Single Shot MultiBox Detector (SSD), YOLO (You Only Look Once), YOLO9000
7	YOLOv3, YOLOv4; graph neural networks, DeepWalk, Word2Vec
8	RetinaNet, YOLOv5 through YOLOv8; SORT, ByteTrack, and BoT-SORT object tracking
9	DeepSORT, SMILEtrack, StrongSORT; vision transformers
10	Review and midterm preparation
11	**Midterm exam**
12	Vision transformers (continued), DETR, Deformable DETR, Swin Transformers, Segmentation Transformer, Segmenter; generative agents; open problems in research
13	Pix2Seq, CLIP, diffusion models, ControlNet
14	OWL-ViT, video latent diffusion models, Segment Anything